perplexity

Executive view

Liquid cooling is moving from a specialist technology to a **standard design option for new AI-oriented data centers**, but adoption across the broader enterprise estate remains limited. The transition is being driven less by ordinary cloud workloads and more by GPU rack density, higher chip thermal design power, power constraints and the need to operate more compute within the same facility.

The clearest current survey signal is that approximately **22% of data centers report using some form of liquid cooling**, while **61% of operators are actively considering it.** However, among facilities that have adopted it, 48% use liquid cooling in fewer than 10% of server racks. [1]

Adoption levels

Segment	Adoption pattern
AI hyperscale campuses	Fastest adoption; direct-to-chip liquid cooling is becoming the default for high-density GPU clusters
Large colocation facilities	Increasing adoption, especially where customers request GPU-ready halls
Enterprise data centers	Still early; adoption is usually limited to AI/HPC pods, research clusters or new halls
Legacy facilities	More difficult because of plumbing, CDU, floor-loading, leak-management and operational changes
Greenfield facilities	Best fit because the building, power, cooling loop and rack layout can be designed together

A useful distinction is between **facility adoption** and **rack penetration**. A data center may technically “use liquid cooling” while only a small AI cluster is liquid-cooled; this is why reported adoption rates can appear high even when most installed racks still rely on air cooling.

Main technologies

Direct-to-chip cooling

Direct-to-chip, or cold-plate cooling, circulates coolant through plates attached to GPUs, CPUs or other high-power components. Heat is transferred from the chip into the liquid loop and then rejected through a coolant-distribution unit, dry cooler, chiller or facility-water system.

This is currently the leading architecture for large AI deployments. Dell’Oro expects **single-phase direct liquid cooling** to remain the dominant approach through the end of the decade. [2]

Rear-door heat exchangers

A rear-door heat exchanger attaches to the back of a rack and removes heat from the server exhaust before it enters the data-center room. It is attractive for brownfield facilities because it can often be introduced without converting every server to direct-to-chip cooling.

It is best viewed as a **transition technology**: useful for medium-to-high rack densities, but less effective than direct-to-chip systems when the majority of rack heat must be removed through liquid.

Immersion cooling

Servers or selected components are immersed in a dielectric fluid. Immersion can provide strong thermal performance and lower fan energy, but it introduces more substantial operational changes:

- Server maintenance and handling are different.
- Fluid management becomes a new facilities responsibility.
- Hardware compatibility must be verified.
- Standardization and resale pathways are less mature.

Immersion is therefore likely to remain selective, particularly for specialized HPC, cryptocurrency, edge or extreme-density deployments. [2]

Two-phase cooling

Two-phase systems use a fluid that boils at the heat source and condenses elsewhere in the loop. They can support high heat flux, but concerns around fluid cost, containment, reliability and environmental regulation have limited broad adoption.

Why adoption is accelerating

1. Rack power density

AI racks are moving beyond the practical comfort zone of conventional air cooling. Traditional enterprise racks may operate at single-digit or low-double-digit kilowatts, while modern GPU systems can move toward **tens or even hundreds of kilowatts per rack**.

As rack density rises, air cooling requires more airflow, larger fans, greater pressure, bigger cooling plants and more floor-space-consuming equipment. Liquid cooling removes heat more efficiently and allows higher compute density.

2. Energy efficiency

Liquid cooling can reduce fan energy and improve the efficiency of heat rejection, although the total facility benefit depends on the design. Pumps, coolant-distribution units, heat exchangers and chillers still consume power.

Market research estimates suggest liquid cooling can reduce cooling-related power consumption by up to 40% in some deployments, but such figures should be treated as design-dependent rather than universal. [3]

3. Capacity constraints

Liquid cooling allows operators to install more compute within an existing electrical envelope and building footprint. This is particularly valuable where:

- Grid capacity is limited.
- New substations take years to build.
- Land is expensive.
- Customers need accelerated GPU deployment.
- The operator wants to increase revenue per megawatt.

This makes liquid cooling part of a broader **capacity strategy**, not merely an HVAC upgrade.

4. Higher chip TDP

Future accelerator generations are expected to continue increasing thermal design power. As heat flux rises at the chip and package level, air cooling becomes increasingly difficult even when facility-level cooling capacity is available.

Main barriers

The IEA 4E/EDNA study identifies the principal barriers as:

- **Lack of standardization:** 39%.
- **High initial cost:** 38%.
- **Long-term reliability concerns:** 35%. [1]

Other barriers include:

- Leak detection and containment.
- Coolant compatibility and material degradation.
- New maintenance procedures.
- Limited technician skills.
- Vendor lock-in around manifolds, CDUs and cold plates.
- Difficulty retrofitting existing halls.
- Insurance and warranty concerns.
- Water quality and corrosion management.
- Uncertainty over server refresh cycles.

The operational risk is often overstated as "water near electronics." In a well-designed direct-to-chip system, the coolant circuit is isolated from the electronics. The more important risks are poor installation, connector failure, inadequate monitoring, contamination and insufficient maintenance discipline.

Brownfield versus greenfield

Factor	Greenfield AI facility	Existing enterprise facility
Rack layout	Designed around liquid-cooled systems	Often constrained by existing racks and aisle design
Power	High-density distribution can be built in	Electrical capacity may be fragmented
Cooling	CDUs, manifolds and heat rejection integrated from start	Retrofit may require major piping and plant changes
Deployment speed	Faster once permits and power are secured	Pilot can be quick, but full conversion is difficult
Best technology	Direct-to-chip	Rear-door heat exchanger or limited direct-to-chip pod
Typical business case	New AI/cloud capacity	Extend life of site or add GPU capability

For enterprises, the most practical pathway is usually not converting the entire data center. It is creating a **liquid-cooled AI zone** or GPU pod within a mixed air-and-liquid facility.

Market outlook

Published market forecasts vary sharply because some reports count only cooling hardware while others include services, heat-rejection systems and broader thermal-management equipment.

- Dell’Oro forecasts the data-center liquid-cooling market approaching **$7B by 2029.** [2]
- A separate market estimate projects growth from approximately **$870M in 2024 to $10.7B by 2030**, implying a CAGR above 50%. [4]
- MarketsandMarkets estimates a much larger long-term market, reaching **$27.65B by 2033**, reflecting a broader market definition. [5]

These figures should not be combined. For a presentation, describe the range as evidence of **rapid growth with substantial definition uncertainty**, not as a precise market size.

Enterprise adoption trend

The enterprise pattern is likely to develop in three stages:

1. **Pilot stage:** liquid-cooled GPU servers in research, financial-services, pharmaceutical, engineering or government workloads.
2. **Pod stage:** dedicated liquid-cooled rooms or rows within colocation and enterprise facilities.
3. **Facility stage:** new data centers designed primarily around high-density liquid-cooled compute.

The industry is currently between stages one and two for most enterprises, while hyperscalers and specialist GPU-cloud providers are already building stage-three facilities.

India-specific relevance

For India, liquid cooling has particular value because it can reduce dependence on large volumes of chilled air in hot climates and support denser deployments where land and grid capacity are constrained. However, the business case must include:

- Water availability and quality.
- Dry-cooler versus evaporative heat-rejection design.
- Monsoon humidity and corrosion control.
- Availability of trained facilities technicians.
- Import dependence for CDUs, cold plates and specialty fluids.
- Local service and spare-parts capability.
- State-level data-center incentives and power tariffs.

India's emerging AI data-center market is likely to adopt a **hybrid model**: air cooling for conventional enterprise and cloud workloads, direct-to-chip liquid cooling for GPU clusters, and rear-door heat exchangers for selected retrofit environments.

Slide-ready conclusion

> **Liquid cooling is not yet universal across enterprise data centers, but it is becoming essential for new high-density AI capacity. Adoption is strongest in greenfield hyperscale and colocation facilities; enterprise uptake will proceed through GPU pods and hybrid air/liquid designs. The decisive drivers are rack density, power availability and facility economics—not sustainability alone.**

For tracking adoption, use these metrics:

- Percentage of facilities using liquid cooling.
- Percentage of racks within those facilities using it.
- Average rack power density in kW/rack.
- Share of new AI capacity using direct-to-chip cooling.
- PUE and cooling-system power reduction.
- Retrofit cost per rack.
- Time to deploy a liquid-cooled GPU pod.
- Coolant-distribution-unit and heat-rejection capacity.

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